

Not every stable cluster is homeostatic: Control, projectibility, and the Boyd–Khalidi divide

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Abstract

Homeostatic property cluster theory gave natural-kinds realism a powerful anti-essentialist form: a kind can be real because its properties cluster through mechanisms rather than because it has an essence (Boyd, 1991, 1999). But the word HOMEOSTATIC has often carried more weight than the theory has earned. Some stable clusters support induction without corrective control, and some Khalidian network nodes support projection through internal order rather than through homeostatic return (Khalidi, 2013, 2018). This paper uses Weinberger’s account of homeostasis as causal control (Weinberger, 2026) to separate three cases: stable property clusters, network-ordered kinds, and strict homeostatic-control kinds. The payoff is a sharper Boyd–Khalidi division of labour: projectibility is the epistemic target, network order explains many projectible kinds, and homeostasis should be reserved for feedback-sensitive maintenance under perturbation.

1 INTRODUCTION

The target is a quiet ambiguity in recent anti-essentialist kind theory. Boyd’s homeostatic property cluster framework, Slater’s stable property cluster proposal, and Khalidi’s causal-network account all reject essences. They also all aim to explain why some categories support projection. But they differ over what does the explanatory work: clustering, stability, internal order, or homeostatic maintenance.

The working claim of this paper is simple: not every stable cluster is homeostatic. Treating it as such makes the framework too permissive. The correction is not to abandon Boyd, but to make HOMEOSTASIS more literal and more demanding by using Weinberger’s causal-control account as the guardrail.

2 THE PROBLEM

Boyd’s framework works because projectibility does not require necessary and sufficient conditions. A category can support induction when its associated properties are held together by mechanisms.

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The risk is that MECHANISM becomes too easy to satisfy. Any recurring stabiliser, any institutional reinforcement, or any reliable co-occurrence can start to look homeostatic.

Khalidi's network account addresses a different weakness. A cluster is more than a list of associated properties. Its properties can be ordered: some generate, sustain, or explain others. That's an advance over a bare cluster picture, but it doesn't by itself recover homeostasis. Ordered dependence is not the same as corrective control.

3 ANALYSIS

The proposed taxonomy has three tiers.

First, stable property clusters support limited induction because their properties co-occur reliably. They may be useful, but their stability does not yet show a corrective mechanism.

Second, network-ordered kinds are projectible because their associated properties occupy structured positions in a causal or mixed network. Khalidi's view is strongest here: order explains why some properties are diagnostic and others derivative.

Third, homeostatic-control kinds are stricter. In these cases, feedback-sensitive mechanisms correct perturbation and preserve a higher-scale relation through lower-scale variation. Here Weinberger's account matters. Homeostasis is not mere constancy, but it is also not mere stability.

4 DISCUSSION

The distinction matters for Brett Reynolds's broader HPC programme in linguistics and social ontology. Linguistic categories may be stable, projectible, and mechanism-maintained without all being homeostatic in the strict control-theoretic sense. Social practices may be stabilized by records, offices, sanctions, training, and uptake without becoming single homeostatic control systems.

The paper should therefore test its taxonomy against two worked cases: one case from biological or engineering control, and one linguistic or social case where stability and projectibility are real but strict homeostasis would overstate the evidence.

5 CONCLUSION

Projectibility remains the point. Mechanisms and network order matter because they explain why projection is reliable. Homeostasis is one important way projection can be secured, but it isn't the name for every stable pattern worth taking seriously.

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REFERENCES

- Boyd, R. (1991). Realism, anti-foundationalism and the enthusiasm for natural kinds. *Philosophical Studies*, 61(1–2), 127–148. <https://doi.org/10.1007/BF00385837>
- Boyd, R. (1999). Homeostasis, species, and higher taxa. In R. A. Wilson (Ed.), *Species: New interdisciplinary essays* (pp. 141–186). MIT Press. <https://doi.org/10.7551/mitpress/6396.003.0012>
- Khalidi, M. A. (2013). *Natural categories and human kinds: Classification in the natural and social sciences*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511998553>

- Khalidi, M. A. (2018). Natural kinds as nodes in causal networks. *Synthese*, 195(4), 1379–1396. <https://doi.org/10.1007/s11229-015-0841-y>
- Weinberger, N. (2026). Homeostasis and causal control. *Biology & Philosophy*, 41, Article 18. <https://doi.org/10.1007/s10539-026-10018-8>